

Name: \_\_\_\_\_



Exit quiz

Construction

Exit quiz

© Oak National Academy 2025. Produced in partnership with Mathematics in Education and Industry.

Licensed on the [Open Government Licence v3.0](#), except where otherwise stated. [See Oak terms & conditions](#).



0

## Construction

---

- 1 1 kW = \_\_\_\_\_ W (Fill in the blank)
- 2 0.96 kW = \_\_\_\_\_ W (Fill in the blank)
- 3  $Q = A \times v$  where  $Q$  is the flow rate,  $A$  is the flow area and  $v$  is the velocity. If the flow rate is  $36 \text{ m}^3$  and the velocity is  $10 \text{ m/s}$ , what is the flow area in  $\text{m}^2$ ?  
\_\_\_\_\_
- 4  $\text{BTU} = \text{volume of room} \times \text{allowance for heat loss}$ . Calculate the BTU for a bedroom with dimensions 3 m by 3 m by 5 m.

| Room        | Allowance |
|-------------|-----------|
| Bathroom    | 121.5     |
| Bedroom     | 108       |
| Dining room | 135       |
| Hallway     | 81        |
| Kitchen     | 81        |
| Living room | 135       |

| Height (mm) | Width (mm) | BTU  |
|-------------|------------|------|
| 600         | 400        | 2184 |
| 600         | 600        | 3280 |
| 600         | 800        | 4372 |
| 600         | 1000       | 5464 |
| 600         | 1200       | 6556 |
| 600         | 1400       | 7649 |
| 600         | 1600       | 8744 |

\_\_\_\_\_

- 5 BTU = volume of room  $\times$  allowance for heat loss. If I want to buy one radiator for my bedroom (dimension 3 m by 3 m by 5 m), which one should I buy? (Tick **1** correct answer)

| Room        | Allowance |
|-------------|-----------|
| Bathroom    | 121.5     |
| Bedroom     | 108       |
| Dining room | 135       |
| Hallway     | 81        |
| Kitchen     | 81        |
| Living room | 135       |

| Height (mm) | Width (mm) | BTU  |
|-------------|------------|------|
| 600         | 400        | 2184 |
| 600         | 600        | 3280 |
| 600         | 800        | 4372 |
| 600         | 1000       | 5464 |
| 600         | 1200       | 6556 |
| 600         | 1400       | 7649 |
| 600         | 1600       | 8744 |

- ☐ 600 mm by 400 mm
- ☐ 600 mm by 600 mm
- ☐ 600 mm by 800 mm
- ☐ 600 mm by 1000 mm
- ☐ 600 mm by 1200 mm

- 6 BTU = volume of room  $\times$  allowance for heat loss. If I want to buy two radiators for my bedroom (dimension 3 m by 3 m by 5 m), which ones should I buy? (Tick **1** correct answer)

| Room        | Allowance |
|-------------|-----------|
| Bathroom    | 121.5     |
| Bedroom     | 108       |
| Dining room | 135       |
| Hallway     | 81        |
| Kitchen     | 81        |
| Living room | 135       |

| Height (mm) | Width (mm) | BTU  |
|-------------|------------|------|
| 600         | 400        | 2184 |
| 600         | 600        | 3280 |
| 600         | 800        | 4372 |
| 600         | 1000       | 5464 |
| 600         | 1200       | 6556 |
| 600         | 1400       | 7649 |
| 600         | 1600       | 8744 |

- ☐ Two 600 mm by 400 mm radiators
- ☐ Two 600 mm by 600 mm radiators
- ☐ Two 600 mm by 800 mm radiators
- ☐ Two 600 mm by 1000 mm radiators
- ☐ Two 600 mm by 1200 mm radiators